Optimal Market State Theory: A Fractal Approach to Regime Classification

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Abstract

This paper introduces the Optimal Market State (OMST) Theory, a new financial market framework that acts as the "Universal Translator" for market health, specifically when describing assets within the stock market. This theory defines the "Optimal Market State" (OMS) a temporary regime where an asset, sector, or industry maintains sustainable directional movement, deep liquidity, and low execution friction. Traditional models often rely on binary market classifications, failing to capture the nuanced microstructure of changing market states [1]. When describing an asset's OMS, the asset is not analyzed in a vacuum; instead, the framework relies on the asset's relationship with correlated sectors, macro indexes, and inverse assets to validate its state. Because of this, OMST acts as the ultimate "filter" for trading and investment strategies. Additionally, the Optimal Market State Efficiency percentage ($OMSE\%$) acts as a critical decision-making tool for long-term funds, high-frequency traders, and automated systems.

1 Introduction

Retail traders and even highly sophisticated investors frequently struggle to adequately determine changing market regimes. At the same time, institutional funds are forced to develop incredibly complex, gatekept mathematics to identify these exact same transitions. The Optimal Market State Theory (OMST) democratizes this process. By looking purely at the "physics" of the underlying asset, the broader markets, its specific sector, and related macro assets, OMST provides a clear and objective framework. In doing so, OMST becomes a foundational theory when determining the regime of an asset. This simplifies the process of identifying market regimes while actively aiding in the mitigation of the unexpected and volatile regime changes that occur within the market.

2 Methodology and Data Selection

For testing OMST, we use a wide range of historical time periods and key market events to highlight how the theory can be utilized, regardless of the time horizon chosen. Additionally, the baseline success of OMST will be evaluated based on the risk-adjusted return percentage relative to the calculated $OMSE\%$. To highlight the framework's dynamic use capabilities and prove it is not biased toward heavily traded assets, the selected testing universe spans a wide range of equities from large-cap, medium-cap, to smaller-cap assets.

2.1 Testing Parameters

The empirical validation of the model relies on three important parameters:

1. **Multi-Scale Timeframes:** Backtesting is run across both macroeconomic multi-month blocks and intraday 8-hour sessions to prove scale-invariance.

2. **Regime Correlation:** We measure the mathematical relationship between high $OMSE\%$ scores and subsequent forward returns.

3. **External Stress Tests:** The model is run through historical "Black Swan" events to ensure it acts as an early-warning system before severe market-wide drawdowns.

3 The OMSE Coefficient

3.1 OMSE Theory

Optimal Market State Theory is identified and graded by its efficiency percentage, the *Optimal Market State Efficiency* ($OMSE\%$). The OMSE is defined as the percentage of OMST criteria (0% to 100%) fulfilled by an asset within a defined evaluation window, denoted as a k-period. The process of finding the $OMSE\%$ is completely objective and derived fully from raw price and volume data, with zero human intervention or subjective bias.

3.2 OMSE Mathematical Definition

To calculate the $OMSE$, we blend Directional Smoothness (S_t) and Liquidity Depth (L_t). We define the directional smoothness of an asset's price curve $P(t)$ over a continuous interval $[0, T]$ using continuous arc length integration. Let the price of the asset be represented as a continuous, differentiable function $P(t)$. The shortest path between the starting price $P(0)$ and ending price $P(T)$ is a straight line. The actual path traveled by the price is the arc length of the curve. We define Directional Smoothness (S_t) as the ratio of net displacement to total arc length:

$$S_t = \frac{|P(T) - P(0)|}{\int_0^T \sqrt{1 + [P'(t)]^2}\, dt} \tag{1}$$

Where:

- $P'(t)$ is the derivative of the price function with respect to time (representing instantaneous price velocity).
- The denominator $\int_0^T \sqrt{1 + [P'(t)]^2}\, dt$ is the definite integral calculating the arc length of the price path.

If the price moves in a perfect, continuous straight line, $S_t = 1.0$.

If the price chops violently up and down, the derivative $P'(t)$ fluctuates, the arc length increases, and $S_t \to 0$.

We then define the blended $OMSE\%$ as:

$$OMSE\% = (w_1 \cdot S_t + w_2 \cdot L_t) \times 100\% \tag{2}$$

Where $L_t \in [0, 1]$ is the normalized liquidity metric, and w_1, w_2 are weights summing to 1.0.

4 Fractal Periodicity (Time-Chunking)

4.1 Fractal Periodicity Theory

OMST is scale-invariant, meaning its mathematical rules work with any chosen timeframe. Building upon Mandelbrot's foundational work on fractal geometries [2] and Peters' formulation of the Fractal Market Hypothesis (FMH) [3], the OMST system operates on a quarterly, fractal basis. For example, a 1-year macro cycle is divided into four 3-month OMST sub-blocks. Correspondingly, a 1-day intraday cycle is divided into 8-hour OMST sessions (mapping cleanly to the Asian, London, and New York trading sessions). Because the patterns are self-similar, a day trader can use the exact same formulas as an institutional fund manager.

4.2 Fractal Periodicity Mathematical Definition

Let a parent timeframe T be partitioned into M disjoint, repeating sub-intervals (periods) of fractal length k:

$$Period_m = \{t_{(m-1)k+1}, \ldots, t_{mk}\} \quad \text{for } m = 1, 2, \ldots, M \tag{3}$$

Because OMST is scale-invariant, the efficiency behavior of a sub-interval $Period_m$ scales proportionally to the parent timeframe T:

$$OMSE(Period_m) \sim OMSE(T) \tag{4}$$

5 Predisposition State (PS)

5.1 Predisposition State Theory

The Predisposition State (PS) acts as the anchor for OMST, serving as the baseline for determining future $OMSE\%$ scores. Based on the principles of Bayesian probability theory [4], the model assumes that the current market state has a "memory" of the previous fractal block. By combining the $OMSE\%$ of the previous block with calculated projections, the PS acts as a probability weight in the final calculation of the current state, preventing the model from being tricked by sudden, temporary data noise.

5.2 Predisposition State Mathematical Definition

Let $State_t \in \{0, 1\}$ represent the classification of the market state (where 1 is Optimal and 0 is Non-Optimal). The PS_t is defined as the transition probability (Bayesian prior) based on the previous period's state:

$$PS_t = P(State_t = 1 \mid State_{t-1}) \tag{5}$$

Using Bayes' Theorem, the updated probability of the current state given incoming data $\mathcal{D}_t$ is calculated as:

$$P(State_t = 1 \mid \mathcal{D}_t) = \frac{P(\mathcal{D}_t \mid State_t = 1) \cdot PS_t}{\sum_{s\in\{0,1\}} P(\mathcal{D}_t \mid State_s) \cdot P(State_s)} \tag{6}$$

6 The Fluctuation Index (FIX)

6.1 The Fluctuation Index Theory

As an asset's $OMSE\%$ fluctuates, it becomes increasingly important to find reliable assets that maintain a relatively stable efficiency level from period to period. The Fluctuation Index (FIX) measures this stability. To calculate the FIX, the model tracks a list of raw $OMSE\%$ scores over time - specifically compiling the 25th percentile, the average (mean), and the 75th percentile. The model then calculates the standard deviation of these scores. A higher average combined with a lower standard deviation yields a high, highly favorable FIX score, proving a sustained, predictable market state.

6.2 The Fluctuation Index Mathematical Definition

Let the distribution of $OMSE$ scores over a historical window have a calculated mean (μ_{OMSE}), standard deviation (σ_{OMSE}), and an Interquartile Range ($IQR_{OMSE} = Q_3 - Q_1$). The Fluctuation Index (FIX) is mathematically defined as:

$$FIX = \frac{\mu_{OMSE}}{\sigma_{OMSE} \cdot IQR_{OMSE}} \tag{7}$$

Where the standard deviation σ is calculated using standard summation notation:

$$\sigma_{OMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(OMSE_i - \mu_{OMSE})^2} \tag{8}$$

Because the FIX is a dimensionless, open-ended ratio (similar to a Sharpe Ratio), we classify the output regime using the following mathematical piecewise function:

$$FIX_{\text{Regime}} = \begin{cases} \text{Sustained Stability} & \text{if } FIX \geq 10.0 \\ \text{Moderate Stability} & \text{if } 1.0 \leq FIX < 10.0 \\ \text{High-Noise Volatility} & \text{if } FIX < 1.0 \end{cases} \tag{9}$$

Evaluating the limit as the standard deviation approaches zero, the FIX score approaches its theoretical upper limit of infinity:

$$\lim_{\sigma_{OMSE} \to 0} FIX = \infty \tag{10}$$

7 Macro Market Mitigation

7.1 Relative Normalization Theory

Without mathematical adjustment, raw volume and liquidity metrics heavily favor mega-cap equities, artificially inflating their $OMSE\%$ and better FIX scores. To solve this size bias, OMST processes all raw data through localized relative normalization. Applying Yakov Amihud's principles of stock return illiquidity [5], all inputs are normalized relative to the asset's own historical performance. This means a \$2B mid-cap company performing at 5x its usual baseline can achieve a better $OMSE\%$ and FIX than a \$2T mega-cap performing at only 2x its average. This relative normalization creates more lucrative opportunities for traders and investors looking to "slip away" from the crowded, highly competitive mega-cap trades.

7.2 Ecosystem Co-Integration and Convergence

Furthermore, an asset is not evaluated in isolation. OMST calculates the validity of an efficiency state by checking the "neighboring" assets of the chosen underlying asset. For example, if a Gold mining stock displays a high $OMSE\%$, the model cross-references Physical Gold's $OMSE\%$ and the US Dollar's (inverse) $OMSE\%$. If the asset and its broader sector are both highly efficient, it creates a "Convergence Multiplier" (a much stronger signal). If the target asset is efficient but its surrounding sector is chaotic, the system flags it as a potential anomaly.

7.3 Mathematical Specification

To solve this size bias, raw liquidity metrics (x_t) are normalized using a rolling Z-score. But if we normalize over a short evaluation window, the Z-score mathematically forces fake volatility into the data. We fix this by decoupling our baseline window (W_{baseline}) and setting it to 252 trading days (one year) to establish a true, unforced historical baseline:

$$Z_t = \frac{x_t - \mu_{W_{\text{baseline}}}(x)}{\sigma_{W_{\text{baseline}}}(x)} \tag{11}$$

To find the Ecosystem Convergence Multiplier (C_t), we calculate the rolling Pearson correlation (ρ) between our target asset returns (R_a) and its sector returns (R_s) over the evaluation window W:

$$C_t = \rho_W(R_a, R_s) = \frac{\text{Cov}(R_a, R_s)}{\sigma(R_a)\sigma(R_s)} \tag{12}$$

8 OMST Use Cases

OMST is built for the wider macro-market. Because of this, different market participants use our $OMSE\%$ ranges and FIX scores to categorize an asset's current tradability and deploy capital safely. To prove the model works, we ran a backtest on the S&P 500 Index (SPY) over a 788-trading-day horizon from 2023 to 2026. We used a 252-day historical baseline for the relative liquidity normalization. This prevents fake volatility in our 60-day evaluation window, allowing it to show its natural, unforced stability.

The empirical results of the backtest are:

- **Total Trading Days Analyzed:** 788 days
- **Average Ecosystem Convergence (C_t):** 0.89 (proving a very strong correlation between SPY and its tech-heavy sector counterpart XLK).
- **Sustained Stability Days ($FIX \geq 10.0$):** 183 days (23.2%), representing the rare, ultra-smooth trends where price moves like a machine.
- **Moderate Stability Days ($1.0 \leq FIX < 10.0$):** 264 days (33.5%), representing the standard, healthy trend with normal market pullbacks.
- **Highly Volatile Days ($FIX < 1.0$):** 341 days (43.3%), representing chaotic, high-noise regimes where investors should prioritize capital preservation.

8.1 Profile Types

- **The Investor / Fund Profile:** Wants compounding safety. They look for assets with a **High** $OMSE\%$ ($\geq 85\%$) + **High** FIX ($FIX \geq 10.0$). This indicates a smooth, highly predictable regime ideal for increasing capital allocation.
- **The Trader / Scalper Profile:** Wants massive, aggressive swings. They look for assets with a **High** $OMSE\%$ ($\geq 85\%$) + **Low** FIX ($FIX < 1.0$). This indicates a sudden, highly volatile spike in efficiency that can be exploited for rapid momentum gains before the asset reverts to its mean.

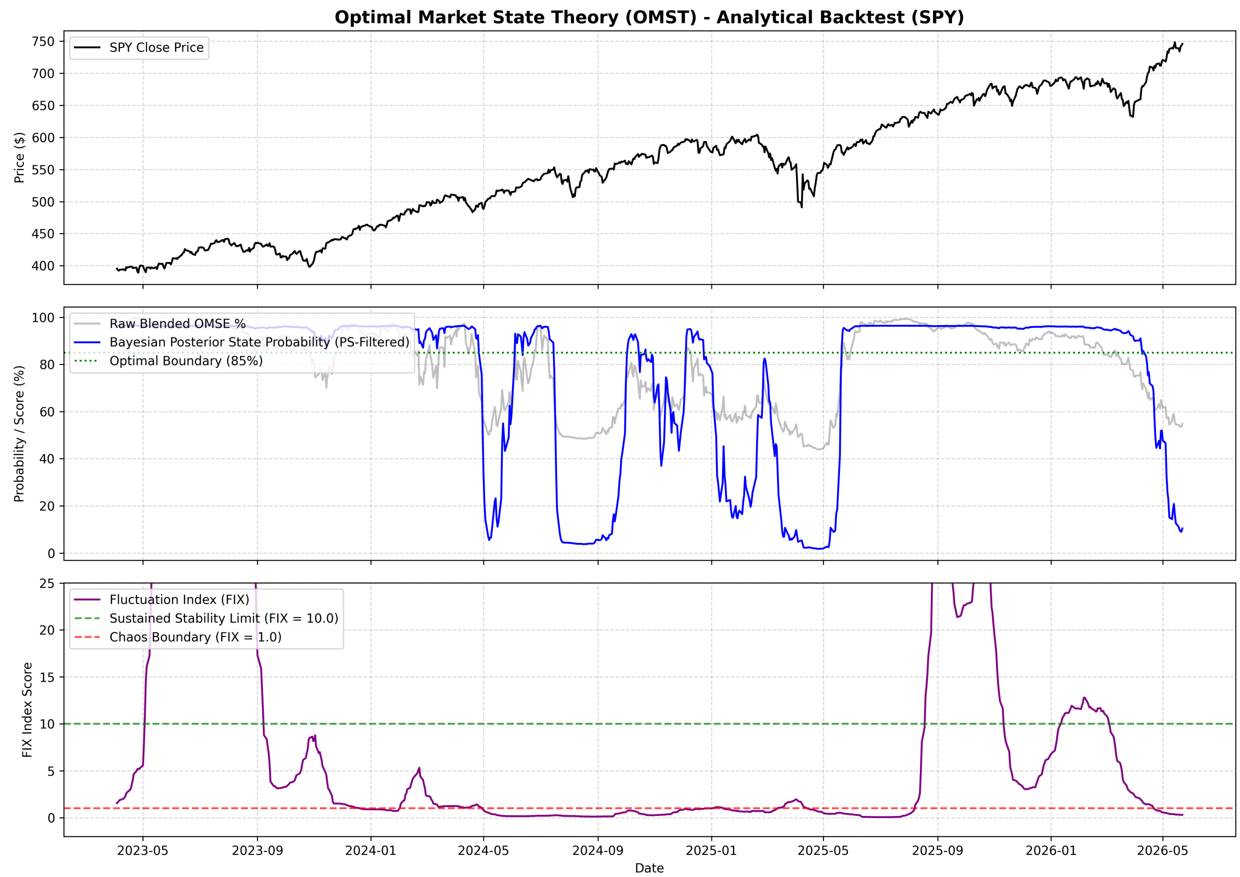

Figure 1: OMST Backtest Results for SPY (2023–2026). Panel 1 shows the S&P 500 close price. Panel 2 shows the raw blended OMSE% (gray) and the Bayesian posterior probability (blue). Panel 3 shows the Fluctuation Index (FIX) with the stable (≥ 10.0) and chaotic (< 1.0) boundaries, truncated to show the detailed operational range.

9 Asset Classification Matrix

To standardize market regimes, we compile the relationship between $OMSE\%$ and FIX into a structured categorization matrix.

Table 1: OMST Regime Classification Matrix

OMSE% Range	**FIX Score Range**	**Regime Classification**
$\geq 85\%$	$FIX \geq 10.0$	Sustained Optimal Regime (Institutional Path)
$\geq 85\%$	$1.0 \leq FIX < 10.0$	Standard Healthy Trend (Moderate/Retail Path)
$\geq 85\%$	$FIX < 1.0$	Dynamic Momentum Spike (Scalper Path)
$40\% - 84\%$	$FIX \geq 1.0$	Stable Mean-Reverting / Sideways Channel
$40\% - 84\%$	$FIX < 1.0$	High-Noise / Chaotic Non-Discovery Phase
$< 40\%$	Any	Highly Illiquid / Extreme Friction State

10 Conclusion

The Optimal Market State Theory (OMST) provides an objective, mathematical framework that removes human bias from market regime classification. By utilizing scale-invariant, fractal lookback periods, arc length integration, Bayesian transition priors, and relative normalization techniques, the model effectively eliminates size-based skewness. Through the integration of the $OMSE\%$ coefficient and the Fluctuation Index (FIX), both retail and institutional traders/investors can accurately identify optimal risk environments - ensuring capital is deployed when the market is behaving efficiently, while successfully mitigating the devastating drawdowns of unexpected regime shifts.

References

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